

Smart Expense Tracker

A Python project using SQLite

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CSE VITYARTHI PROJECT

Problem and Objectives

Problem

Students record expenses in notes, chats and bank apps. There is no single view, so budgets are exceeded without notice.

Objectives

- Record expenses quickly with validated input
- Warn the user before a budget is exceeded
- Show reports, a monthly trend and a simple forecast

Requirements

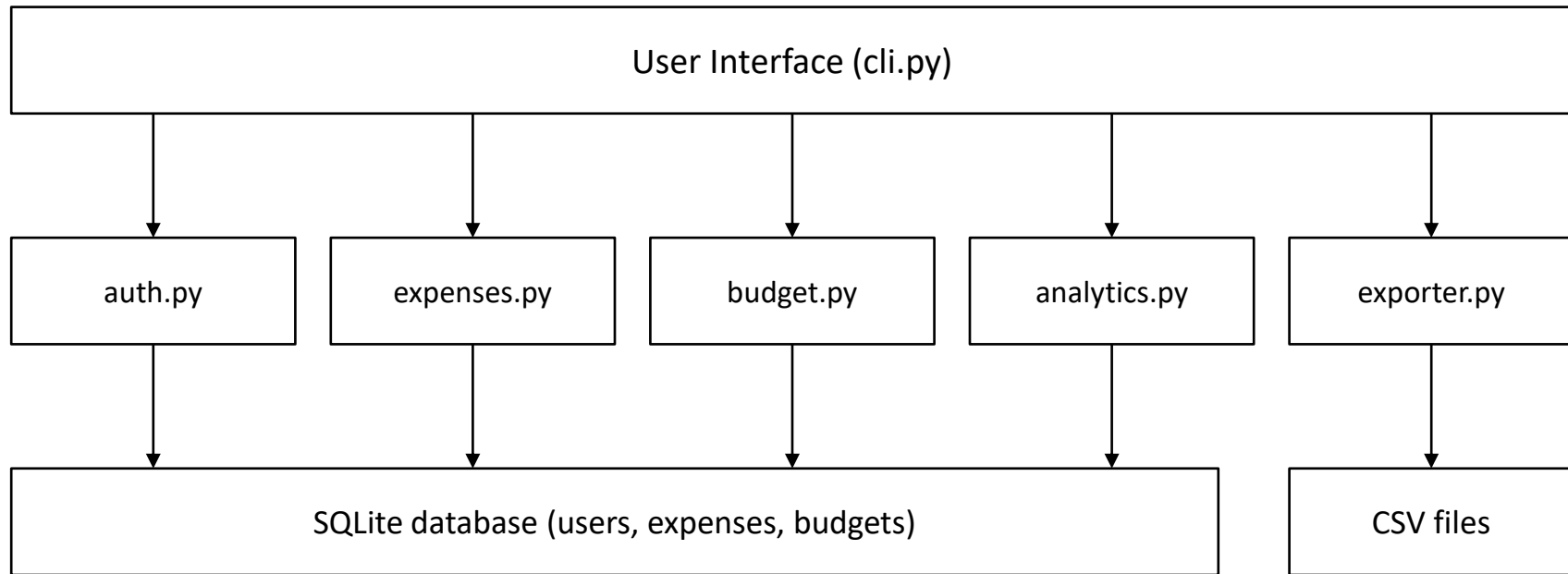
Functional

- User registration and login
- Add, edit, delete, search expenses
- Monthly budgets with alerts
- Reports and forecast
- CSV import and export

Non-functional

- Security: hashed passwords, safe SQL
- Reliability: input validation, error handling
- Maintainability: 12 modules, 31 tests
- Logging: events written to a log file

System Architecture



validators, models, config, exceptions and logging are shared by all modules.

Workflow: Adding an Expense

1. User enters amount, category and date
2. Input is validated
3. Expense is saved in the database
4. Budget is checked
5. Message and alert are shown

Database tables

users (id, username, password_hash, salt) | expenses (id, user_id, amount, category, date) | budgets (user_id, category, month, limit)

Output and Testing

=== Report for 2025-01 ===

Total spent : 12,719.50

Daily average : 1,589.94

Rent (62.9%) 8,000.00

Food (12.3%) 1,570.00

Transport (9.4%) 1,200.00

Education (7.9%) 999.00

Forecast for next month: 4,239.83

- 31 unit tests, all passing
- Tests use an in-memory database
- Covers validation, login, CRUD, budgets, analytics and CSV

Conclusion

Challenges

Safe dynamic sorting and updating, month calculation across years, handling invalid CSV rows.

Learnings

Layered design, input validation, unit testing, using Git.

Future work

Graphical or web interface, charts, recurring expenses.